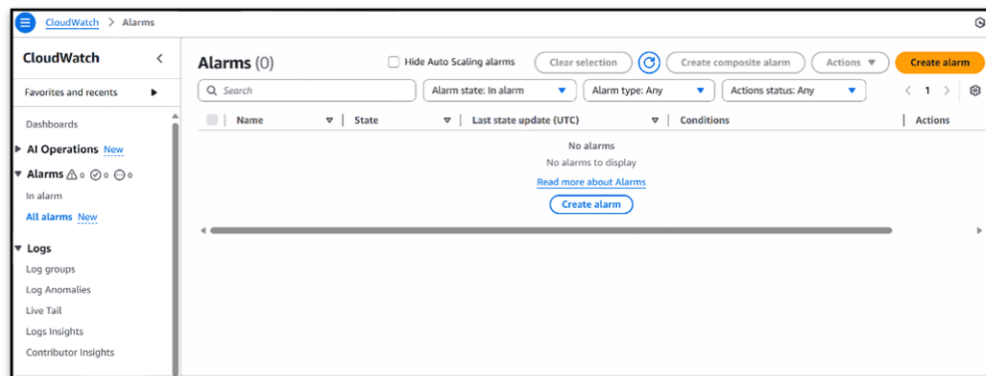


ASSIGNMENT 1

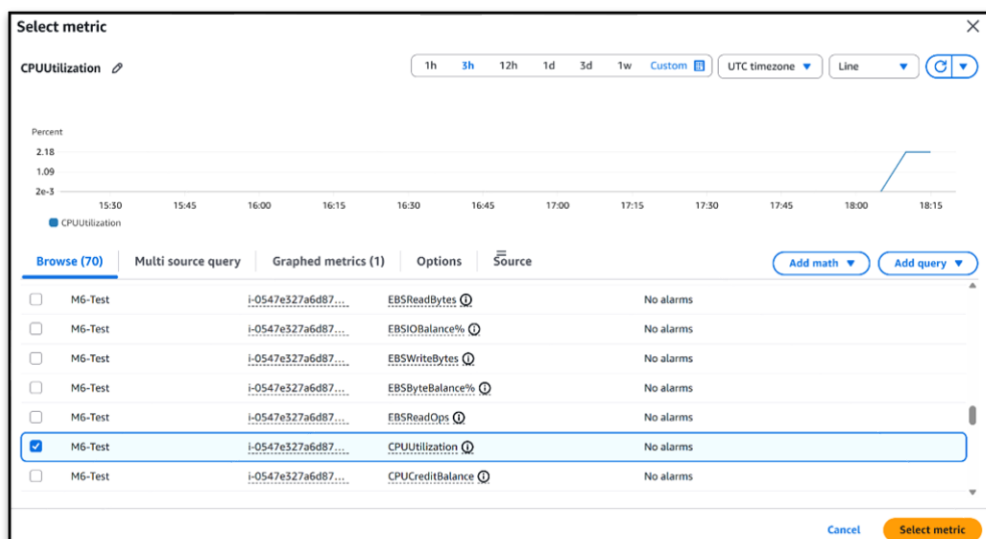
Task 1:

To create a CloudWatch alarm that monitors the CloudWatch metrics for the instance which automatically sends notification when the threshold value is reached and to create an alarm to stop and to terminate an idle instance.

- Open the AWS console → Navigate to CloudWatch services → Click on 'Alarms' on the left pane window → Create alarm.



- Specify metric and conditions: Select metric → EC2 → Pre-instance metrics → Select the Instance which has to be monitored by finding the appropriate metrics (What to be monitored) → Select metric → Specify the period (minutes until alarm triggers) → Use static to specify the threshold → Define alarm condition → specify a threshold value → Next.



CloudWatch > Alarms > Create alarm

Period: 5 minutes

Conditions

Threshold type: ☒ Static (Use a value as a threshold) ☐ Anomaly detection (Use a band as a threshold)

Whenever CPUUtilization is... Define the alarm condition:

☐ Greater > threshold ☐ Greater/Equal >= threshold ☒ Lower/Equal <= threshold ☐ Lower < threshold

than... Define the threshold value: 2.9 (Must be a number.)

▶ Additional configuration

Cancel Next

- **Configure actions:** Select the alarm state on when it is to be triggered (outside threshold, within threshold, just started) → Select an existing SNS topic or create a topic → Specify the email to receive notifications.

CloudWatch > Alarms > Create alarm

Step 1: Specify metric and conditions
Step 2: **Configure actions**
Step 3: Add alarm details
Step 4: Preview and create

Configure actions

Notification

Alarm state trigger: Define the alarm state that will trigger this action.
☒ In alarm (The metric or expression is outside of the defined threshold.)
☐ OK (The metric or expression is within the defined threshold.)
☐ Insufficient data (The alarm has just started or not enough data is available.)

Send a notification to the following SNS topic: Define the SNS (Simple Notification Service) topic that will receive the notification.
☒ Select an existing SNS topic
☐ Create new topic
☐ Use topic ARN to notify other accounts

Send a notification to...: Default_CloudWatch_Alarms_Topic

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints): nareshramesh05@gmail.com - View in SNS Console

Add notification

- **Add EC2 action** (To create an alarm to stop and to terminate an idle instance) → In alarm state (outside the threshold value) → Choose stop instance (or) terminate instance (if needed both by adding another alarm) → Next.

CloudWatch > Alarms > Create alarm

EC2 action

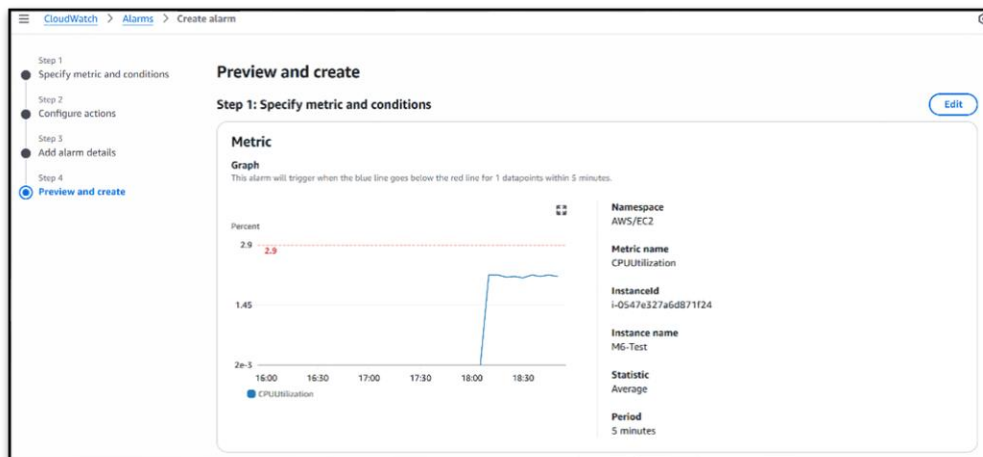
Alarm state trigger: Define the alarm state that will trigger this action.
☒ In alarm (The metric or expression is outside of the defined threshold.)
☐ OK (The metric or expression is within the defined threshold.)
☐ Insufficient data (The alarm has just started or not enough data is available.)

Take the following action...: Define what will happen to the EC2 instance with the Instance ID I-0547e327a6d871f24 when this alarm is triggered.
☐ Recover this instance
☒ Stop this instance
☐ Terminate this instance
☐ Reboot this instance

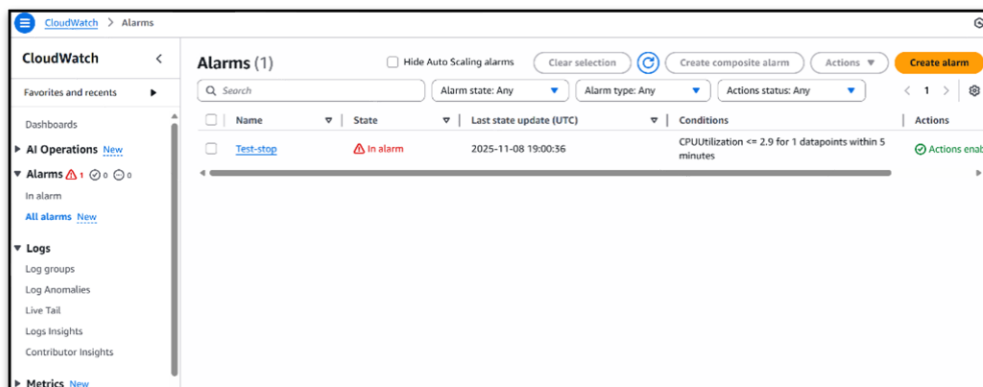
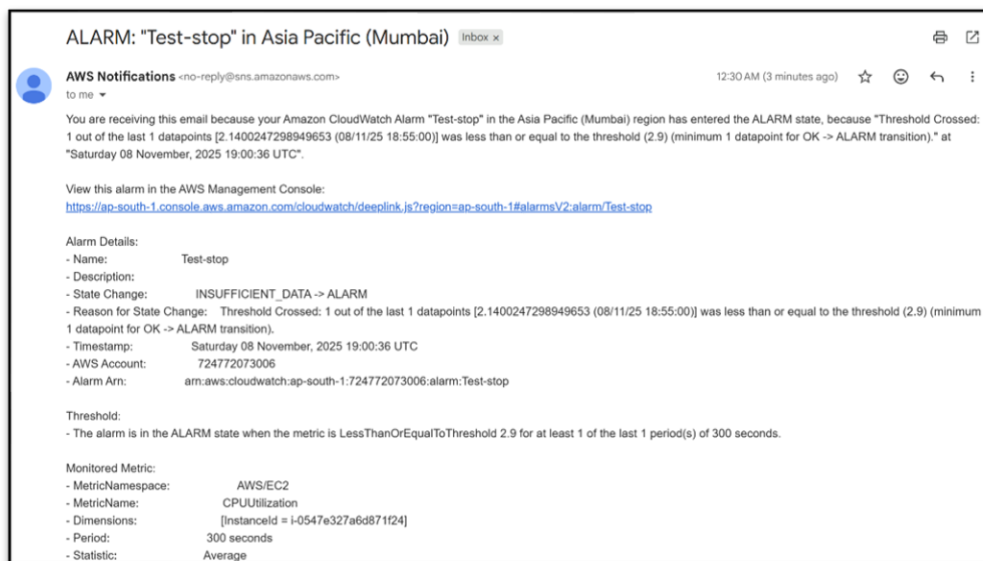
Add EC2 action

Systems Manager action [Learn more](#)

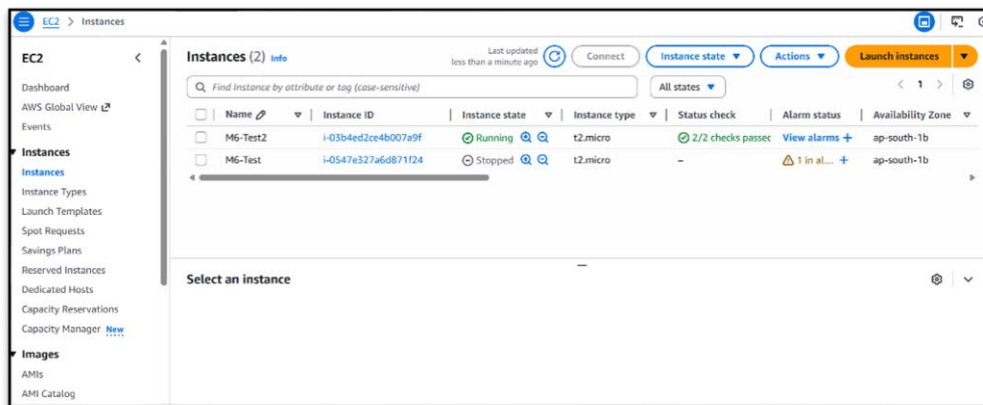
- Add alarm details: Alarm name→Next→Preview and create the alarm.



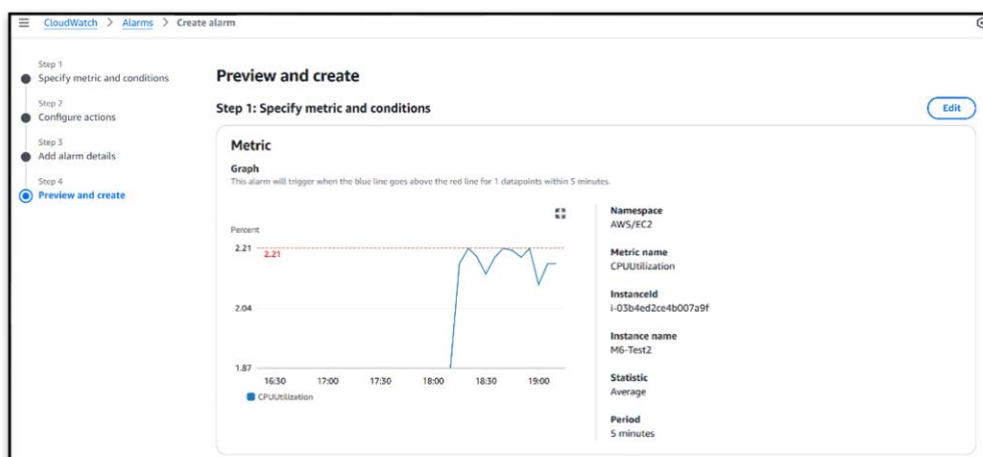
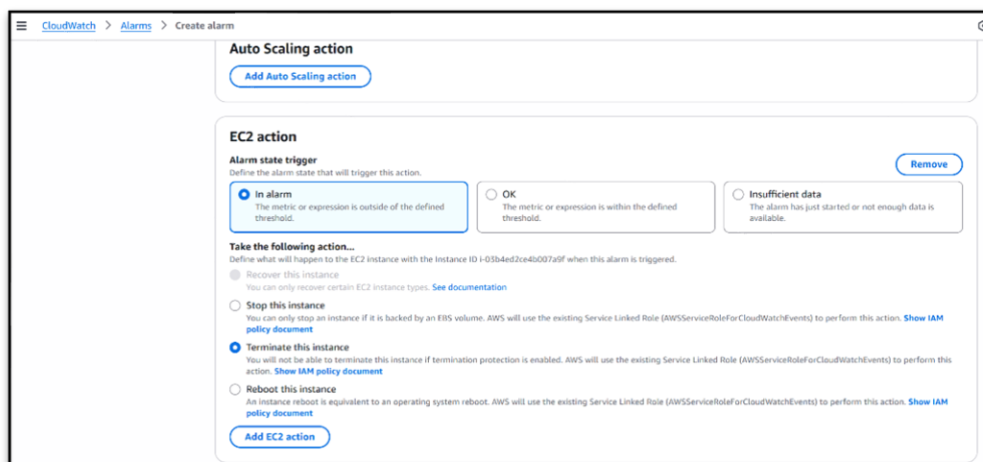
- The alarm has been created.
- You can see that a mail has been sent by the Sns.amazon as the metrics has reached the Alarm state. You can also observe that the status of the alarm is on 'In state' on the AWS console.



- It can also be observed that the instance is on 'stopped' state.



- To terminate an idle instance follow the same steps to set up the alarm as above→Actions→Select terminate instance→Next→Enter the name for alarm→Preview and create.



- The alarm has been created to terminate the instance if the metrics condition is met.

- It can be observed that the instance has been terminated as the metrics reached the threshold value.

